EXPERIMENT NO. 5

AIM: To create SQL views for department salary summary and employee hierarchy, test their updatability, and implement a recursive CTE to display employee reporting chains.

SOURCE CODE:

```
USE companydb;

-- Department Salary View
CREATE VIEW dept_salary_summary AS
SELECT d.dept_id, d.dept_name,
      COUNT(e.emp_id) AS total_employees,
      AVG(e.salary) AS average_salary
FROM Department d LEFT JOIN Employee e
ON d.dept_id=e.dept_id
GROUP BY d.dept_id,d.dept_name;

SELECT * FROM dept_salary_summary;
```

dept_id	dept_name	total_employees	average_salary
1	IT	6	57166.666667
2	HR	6	55833.333333
3	Finance	6	60000.000000
4	Marketing	6	57166.666667
5	AI	6	71000.000000

```
-- Employee Hierarchy View
CREATE VIEW employee_hierarchy AS
SELECT e.emp_id, e.emp_name, e.manager_id,
      m.emp_name AS manager_name
FROM Employee e LEFT JOIN Employee m
ON e.manager_id=m.emp_id;

SELECT * FROM employee_hierarchy;
```

emp_id	emp_name	manager_id	manager_name
1	Aarav	null	null
2	Ananya	1	Aarav
3	Riya	1	Aarav
4	Karan	1	Aarav
5	Neha	2	Ananya
6	Rahul	2	Ananya
7	Ishita	null	null
8	Aditya	7	Ishita
9	Priya	7	Ishita
10	Rohan	7	Ishita
11	Sneha	8	Aditya
12	Vivek	8	Aditya
13	Meera	null	null
14	Arjun	13	Meera
16	Varun	13	Meera
17	Kriti	14	Arjun
18	Yash	14	Arjun
19	Tanya	null	null
20	Dev	19	Tanya
21	Pooja	19	Tanya
22	Manav	19	Tanya
23	Simran	20	Dev
24	Kabir	20	Dev
25	Aditi	null	null
26	Nakul	25	Aditi
27	Ira	25	Aditi
28	Aryan	25	Aditi
29	Diya	26	Nakul
30	Mohit	26	Nakul

```
-- Test view updatability
UPDATE employee_hierarchy
SET emp_name='Aarav Sharma'
WHERE emp_id=1;

-- Recursive CTE
WITH RECURSIVE reporting_chain AS (
    SELECT emp_id, emp_name, manager_id, 1 AS level
    FROM Employee
    WHERE manager_id IS NULL

    UNION ALL

    SELECT e.emp_id,e.emp_name,e.manager_id,rc.level+1
    FROM Employee e
    JOIN reporting_chain rc
    ON e.manager_id=rc.emp_id
)
SELECT * FROM reporting_chain
ORDER BY level, emp_id;
```

emp_id	emp_name	manager_id	level
1	Aarav	null	1
7	Ishita	null	1
13	Meera	null	1
19	Tanya	null	1
25	Aditi	null	1
2	Ananya	1	2
3	Riya	1	2
4	Karan	1	2
8	Aditya	7	2
9	Priya	7	2
10	Rohan	7	2
14	Arjun	13	2
15	Nisha	13	2
16	Varun	13	2
21	Pooja	19	2
22	Manav	19	2
26	Nakul	25	2
27	Ira	25	2
28	Aryan	25	2
5	Neha	2	3
6	Rahul	2	3
11	Sneha	8	3
12	Vivek	8	3
17	Kriti	14	3
18	Yash	14	3
23	Simran	20	3
24	Kabir	20	3
29	Diya	26	3
30	Mohit	26	3